OCNMS Dive Survey Annual Report

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Introduction

Divers from the Northwest Fisheries Science Center (NWFSC) and the National Marine Sanctuaries (NMS) have conducted kelp-forest dive surveys in Olympic Coast National Marine Sanctuary (OCNMS) since 2015 (with the exception of 2020 due to COVID restrictions). Divers from the University of Washington and California State University Monterey Bay have participated as reciprocity divers. Surveys are completed at the end of July to beginning of August. In 2026, surveys were competed between 2026-07-30 and 2026-08-04.

In 2026, divers completed 72 fish transects, 4560 invertebrate transects, and 1020 across five (5) sites (Fig. 1). Divers recorded 23, 47, and 14 taxa of fish, invertebrates, and algae along these transects. To date (2015-2026), the survey has recorded 41, 76, and 17 taxa for fish, invertebrates, and algae.

Methods

Divers on SCUBA conducted in situ surveys to count fish at each site along benthic belt transects (30 m by 2 m) following procedures modified from @malone2022large. Transects were conducted within or directly adjacent to canopy kelp beds (consisting of giant Macrocystis pyrifera or bull Nereocystis luetkeana kelps). In 2015 surveyed at 10 sites and conducted four (4) transects per site at 5 m depth (Fig. 1, [@shelton2018predictable]). From 2016 on, we surveyed at five (5) sites (Fig. 1), sampling at two (2) locations within each site separated by >100 m, and 2 depths within each location (5 and 10 m) Our goal was to complete six (6) replicate transects at each year-site-depth combination [@tolimieri2023changes].

During each fish transect, we counted and estimated the size (total length to nearest cm) of all fishes >5 cm total length; the exception was rockfishes Sebastes spp., for which we estimated sizes of all individuals. Rockfishes $\leq$ 10 cm were considered young-of-year. Divers also estimated horizontal visibility on each transect by determining the distance at which the

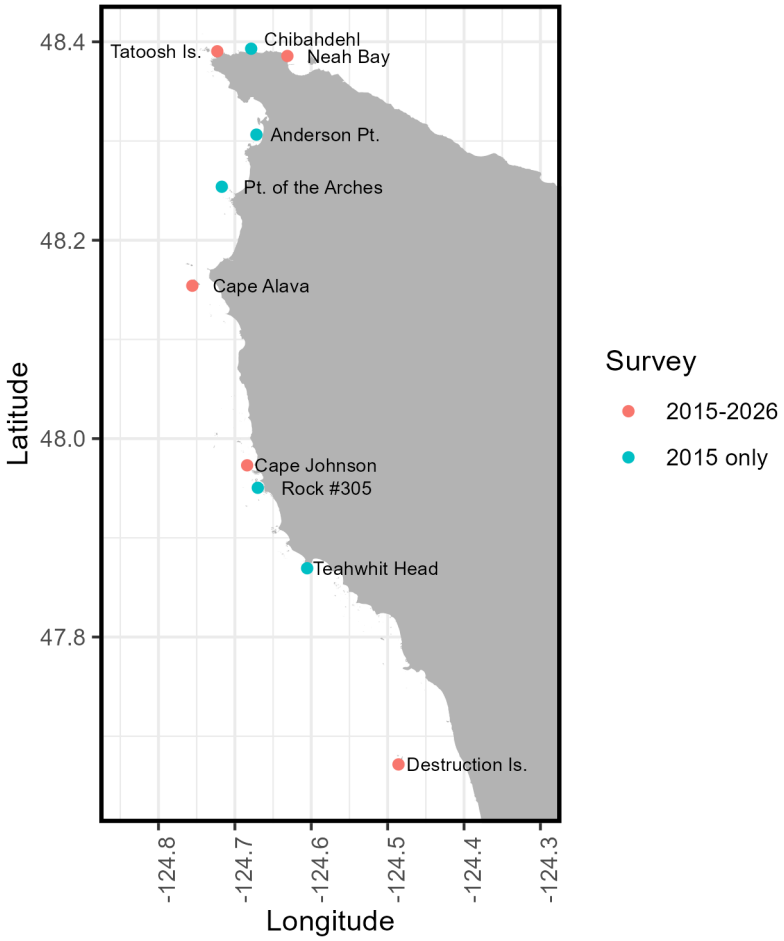


Figure 1: Dive survey locations along the coast of Washington state.

lead diver could distinguish their buddy's extended fingers. Transects with visibility less than 2 m were excluded from analyses.

Results or some other title

Fish

Table 1: Number of fish transects completed by year across 10 sites in 2015 and five sites 2016 to the current year.

Site	2015	2016	2017	2018	2019	2021	2022	2023
Teahwhit Head	4							
Cape Johnson	4	12	16	15	15	16	4	17
Rock 305	4							
Cape Alava	4	12	19	16	16	14	15	16
Point of the Arches	4							
Anderson Point	4							
Tatoosh Island	4	8	13	15	14	14	14	16
Chibadehl Rocks	4							
Neah Bay	4	10	16	15	15	16	13	17
Destruction Island		10	14	15	16	14	4	15

Table 2: Number of fish taxa observed by year across 10 sites

Site	2015	2016	2017	2018	2019	2021	2022	2023
Teahwhit Head	6							
Rock 305	1							
Cape Alava	5	11	10	10	10	11	5	14
Point of the Arches	6							
Anderson Point	1							
Tatoosh Island	4	9	6	10	13	6	5	6

Site	2015	2016	2017	2018	2019	2021	2022	2023
Chibadehl Rocks	3							
Neah Bay	9	10	11	11	16	12	10	12
Destruction Island		6	11	9	12	16		11
Cape Johnson		7	8	9	11	12		10

Adult fishes

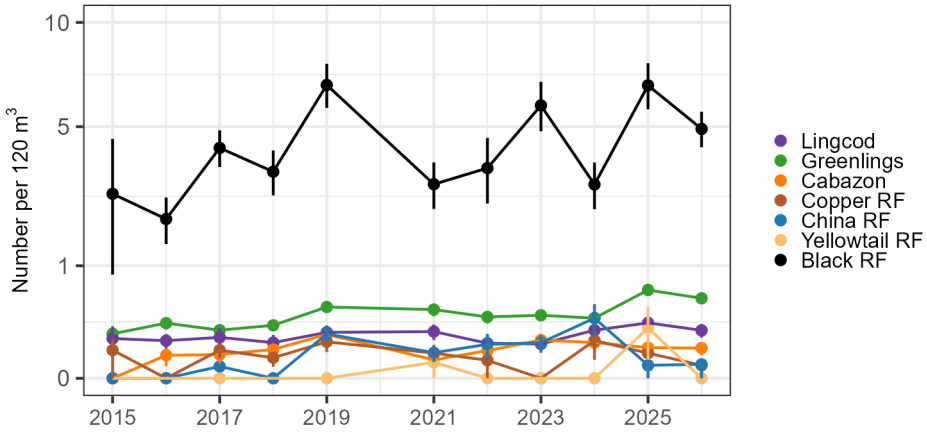


Figure 2: Number of fish per 120 m^3 for 2015-2026 coastwide. Note the scale of the y-axis.

Rockfish young-of-year

It is difficult to visually distinguish many rockfish species when they are small. Therefore, on our surveys, we categorized juvenile rockfishes into five (5) groups established in the literature [@johansson2018seasonal; @markel2020contrasting]:

(1) Yellowtail and black (YTB) included both yellowtail (S. flavidus) and black (S. melanops) rockfishes

(2) The copper/quillback/brown (CQB) group included copper (S. caurinus), quillback (S. maliger), and brown (S. auriculatus) rockfishes

(3) Canary (S. pinniger)

(4) Blue rockfish (S. mystinus)

(5) Unidentified individuals were categorized as juvenile rockfishes

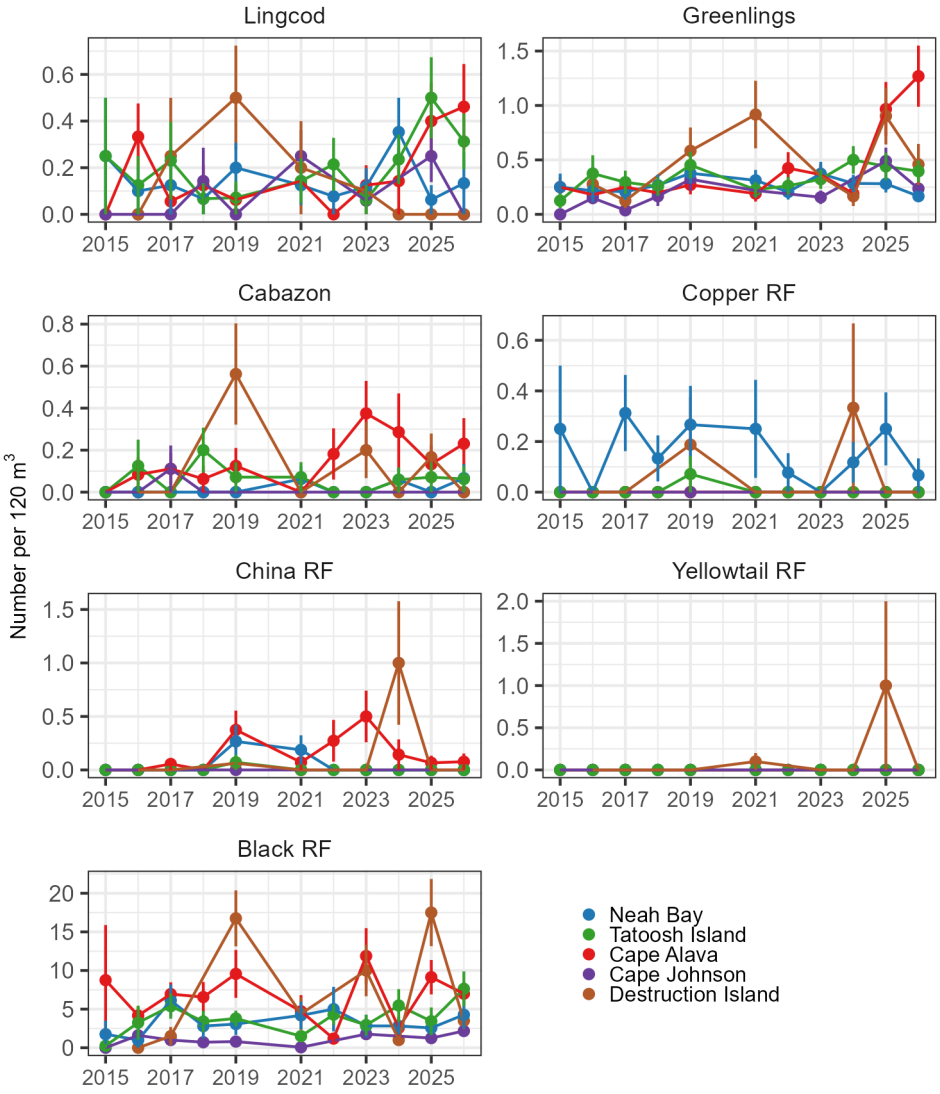


Figure 3: Number of fish per 120 m^3 for 2015-2026 by site.

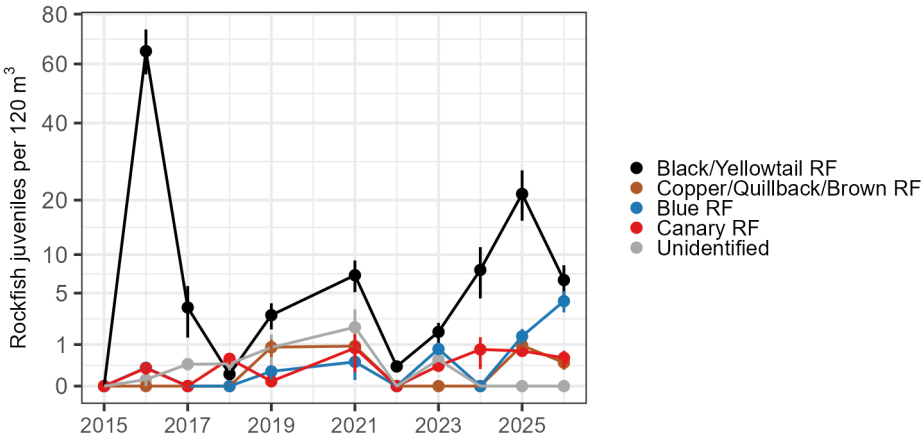


Figure 4: Number of rockfish young-of-year from 2015-2026 coastwide.

Invertebrates

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Urchin trends

Seastars

Pycnopodia

Algae

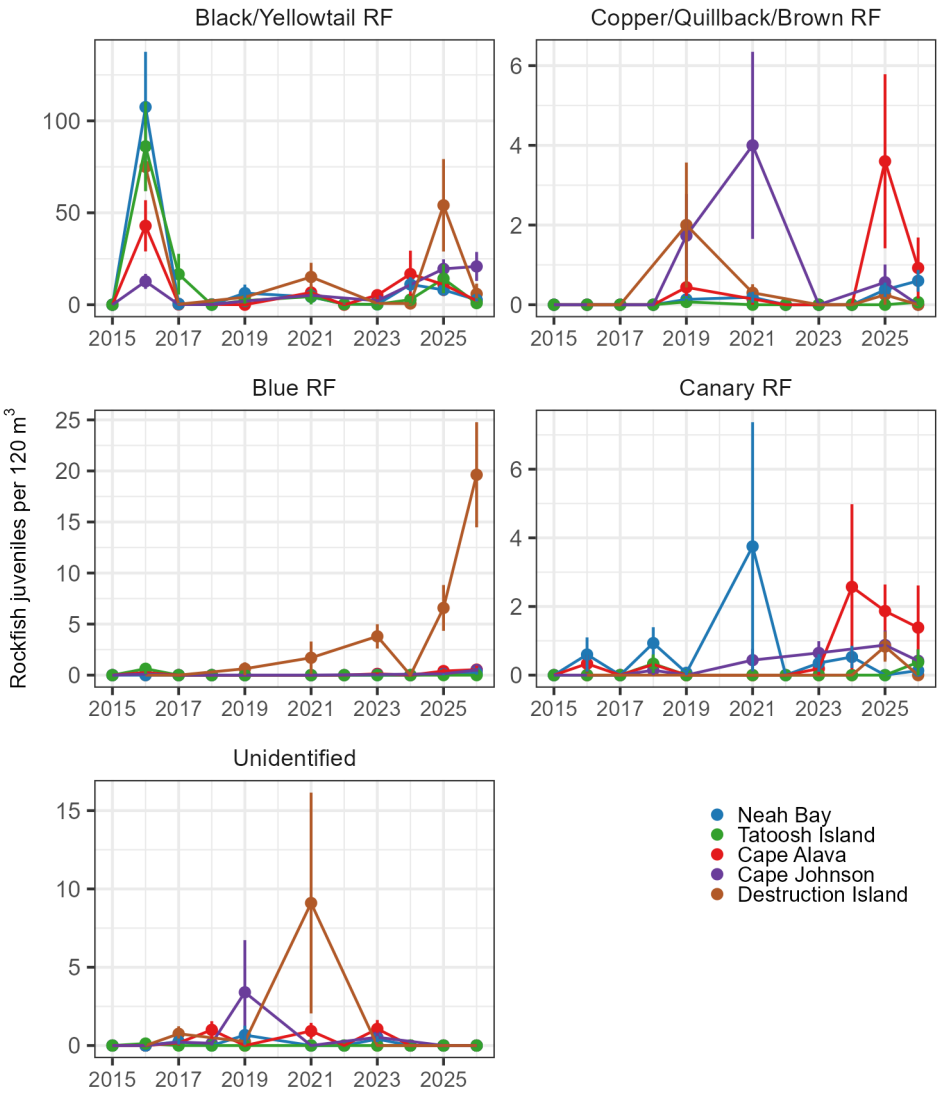


Figure 5: Number of rockfish young-of-year from 2015-2026 coastwide.

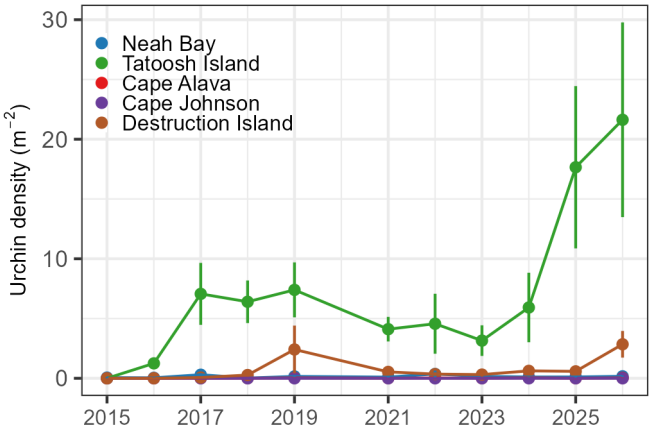


Figure 6

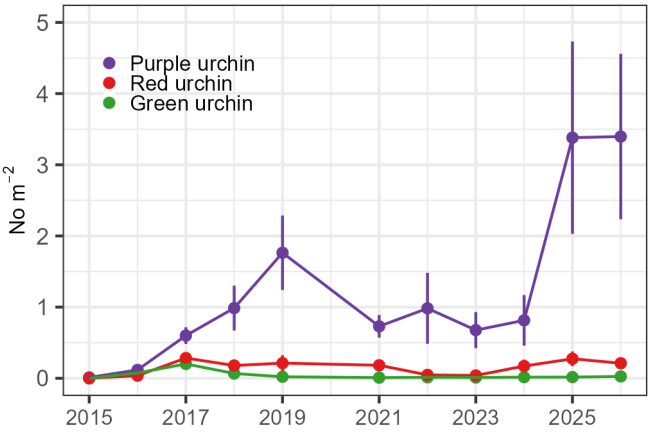


Figure 7

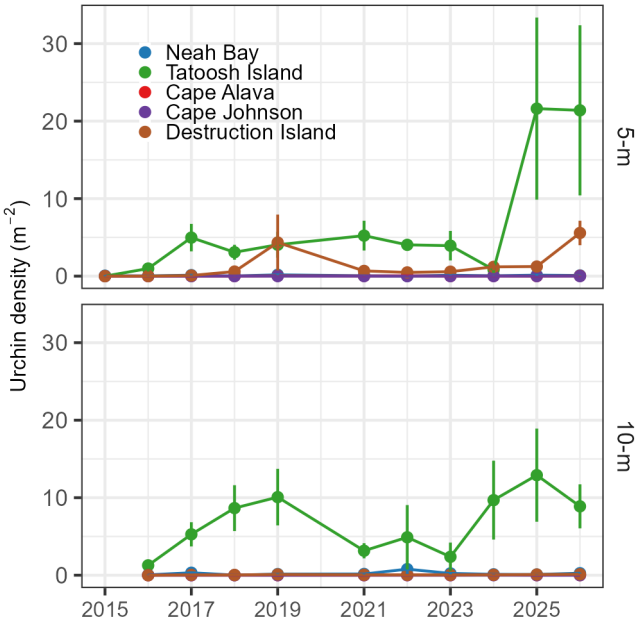


Figure 8

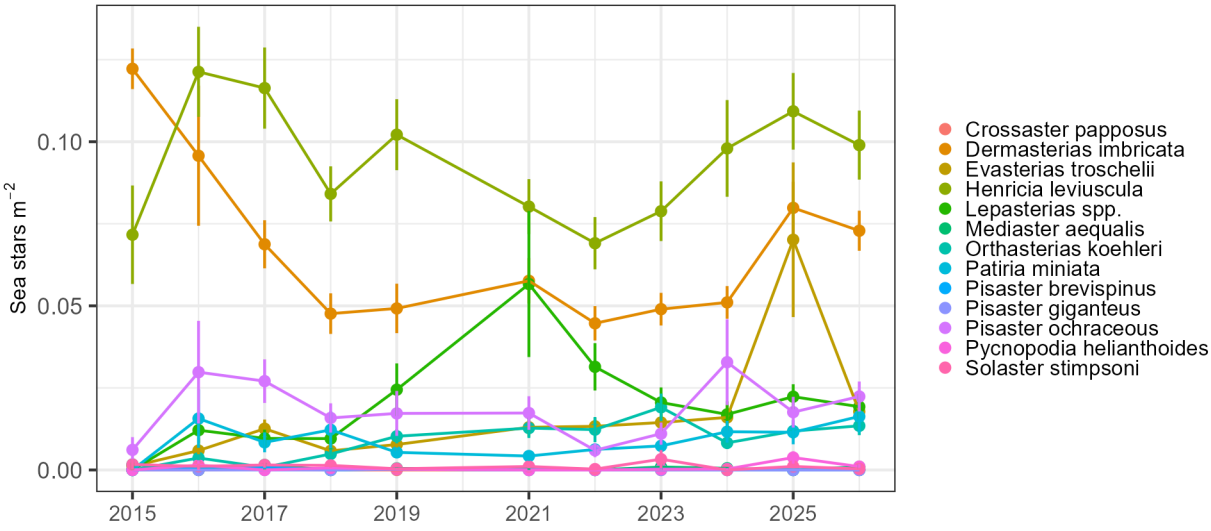


Figure 9

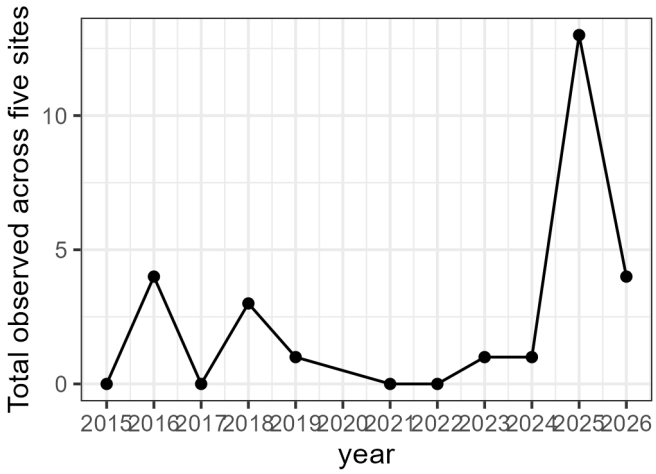


Figure 10

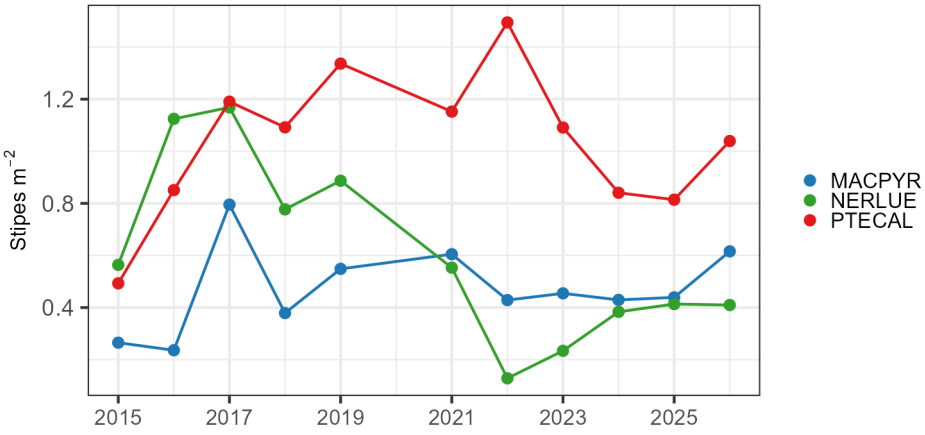


Figure 11

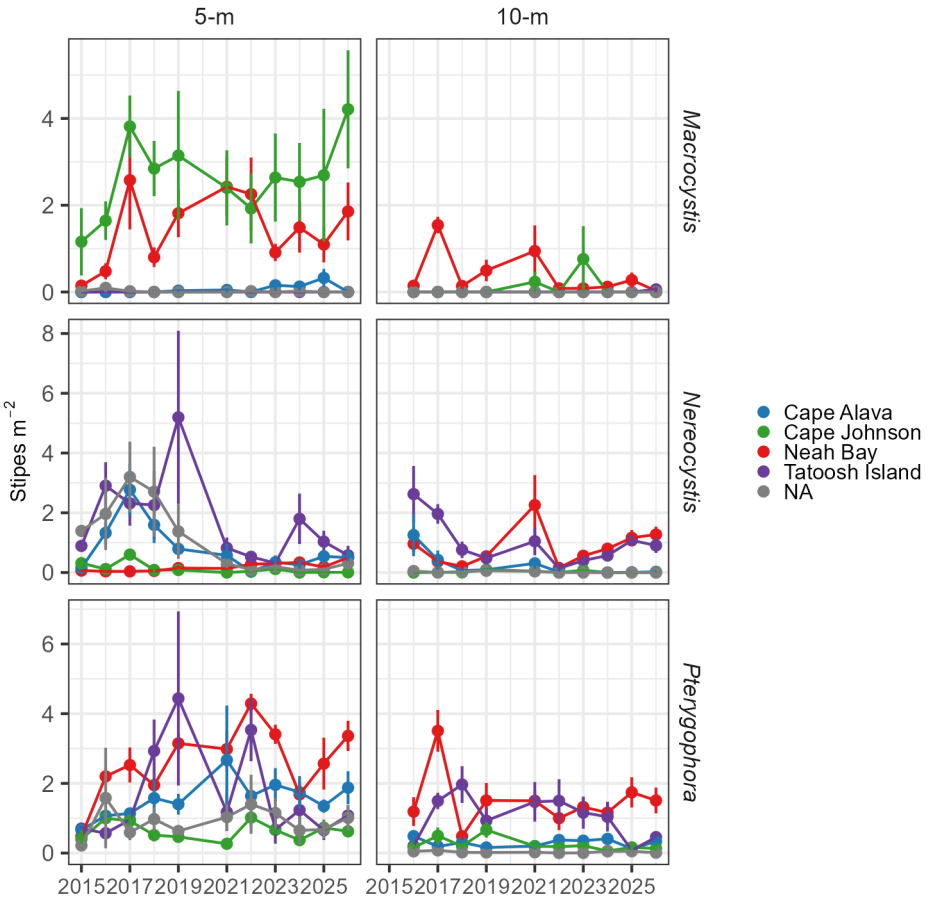

5-m
10-m
Macrocystis
Nereocystis
Pterygophora
Stipes m^{-2}
0
2
4
6
8
2015
2017
2019
2021
2023
2025
Cape Alava
Cape Johnson
Neah Bay
Tatoosh Island
NA

Figure 12